BSDS Statistical Learning I
Problem Set (Updated)

October 28, 2026

1. (a) Based on the following data set, find the product-moment correlation coefficient between $\log(X)$ and $\log(Y)$. [Hint: here the product of X and Y is always 144].

x	1	2	3	4	6	8	9	12
y	144	72	48	36	24	18	16	12

   (b) For a bivariate data set, when do the two types of regression lines: (i) Y on X and (ii) X on Y become identical?

   (c) Suppose that the equations of the two regression lines for a bivariate dataset are $2X+3Y = 17$ and $3X+4Y = 24$. Find the mean of X, the mean of Y, $\text{corr}(X,Y)$ and $\text{var}(X)/\text{var}(Y)$.

2. (a) Consider the following data set consisting of 10 bivariate observations

x	0	1	2	3	4	5	6	7	8	9
y	1	3	5	8	11	13	17	19	23	30

   Find the value of β that minimizes $\sum_{i=1}^{10} |y_i - \beta x_i|$. Comment on the uniqueness of this minimizer.

   (b) Consider n bivariate observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, where $x_i = 2^{i-1}$ for $i = 1, 2, \ldots, n$. Show that the minimizer of $\Delta(\beta) = \sum_{i=1}^{n} |y_i - \beta x_i|$ is y_n/x_n.

   (c) If the mean and median of 10 univariate observations $x_1, x_2, \ldots, x_{10}$ are 2 and 3, respectively, show that the minimizer of $\psi(\theta) = \sum_{i=1}^{10} \left[2(x_i - \theta)^2 + 3|x_i - \theta|\right]$ cannot lie outside the closed interval $[2, 3]$.

3. Suppose that the correlation matrix of a random vector (X, Y, Z) is given by

$$\Sigma_0 = \begin{bmatrix} 1 & \rho & \rho \\ \rho & 1 & \rho \\ \rho & \rho & 1 \end{bmatrix}.$$

   (a) Show that Σ_0 is a positive semi-definite matrix.

   (b) Show that ρ cannot be smaller than -0.5.

   (c) Assume that $\rho = -0.5$. Give a suitable choice of $\mathbf{a} = (a_1, a_2, a_3)$ ($\mathbf{a} \neq \mathbf{0}$) such that variance of $a_1X + a_2Y + a_3Z$ is minimum.

4. (a) Show that in a simple linear regression problem, the product-moment correlation coefficient between the observed and estimated values of the response can never be negative. Also, show that the variance of the estimated values can never exceed the variance of the observed values.

(b) Consider a regression problem where the data set contains 100 bivariate observations, but X takes only two distinct values 1 and 2. Which of the two models, (i) $Y = \beta_0 + \beta_1 X$ or (ii) $Y = \alpha_0 X + \alpha_1 X^2$, will give a better fit (in terms of residual sum of squares) in this case? Justify your answer.

5. Show that in the case of multiple linear regression, $\hat{Y}$, the vector of estimated values of the response, can be expressed as $\hat{Y} = P_x Y$, where Y is the vector of observed values of the response, and P_x is a matrix (known as the projection matrix) that satisfies the following properties.

   (a) $P_x^2 = P_x$, $(\mathbf{I} - P_x)^2 = \mathbf{I} - P_x$ and $(\mathbf{I} - P_x)P_x = 0$.

   (b) Rank of P_x is the same as the rank of the data matrix $\mathbf{X}$.

   (c) Rank(P_x) + Rank$(\mathbf{I} - P_x)$=n.

   (d) $\hat{Y}^\top \hat{Y} = Y^\top P_x Y$ and $e^\top e = (Y - \hat{Y})^\top (Y - \hat{Y}) = Y^\top (\mathbf{I} - P_x) Y$.

   (e) For any vector v lying in the column space of $\mathbf{X}$, $P_x v = v$ and $(\mathbf{I} - P_x)v = 0$.

   (f) If Y belongs to the column space of the data matrix $\mathbf{X}$, then $\hat{Y} = Y$.

   (g) If Y is regressed on $\hat{Y}$, the least square fit will the 45° line passing through the origin.

6. In the case of multiple linear regression, if $\hat{\beta}$ denotes the least squares estimate of β, show that for any other β_0, we have

$$\|Y - \mathbf{X}\beta_0\|^2 = \|Y - \mathbf{X}\hat{\beta}\|^2 + (\beta_0 - \hat{\beta})^\top (\mathbf{X}^\top \mathbf{X})(\beta_0 - \hat{\beta}).$$

7. (a) Show that in the case of simple linear regression (regression of Y on X), the solution of the normal equations (the equations you get by taking the derivative with respect to the parameters and equating them with 0) always exists unless all observed values of X are equal.

   (b) In a simple linear regression problem, can the two regression lines be (i) identical (ii) $2x + 3y = 7$ and $4x - 6y = 2$, (iii) $2x + 3y = 7$ and $4x + 6y = 15$? Give justification to your answer.

8. Consider a regression problem based on observations on three variables X_1, X_2 and Y. Assume that the observations on X_1, X_2 and Y are centered so that you can ignore the intercept term while fitting the regression model. Also assume that the corr(X_1, X_2), corr(X_1, Y) and corr(X_2, Y) are all positive.

   (a) Show that if you regress on Y on X_1, the regression coefficient β_{yx_1} turns out to be positive. Similarly, one can show that if you regress on Y on X_2, the corresponding coefficient β_{yx_2} turns out to be positive as well.

   (b) If you regress Y on X_1 and X_2 simultaneously (multiple linear regression), can any of the estimated regression coefficients be negative? Justify your answer.

   (c) Let the residual sum of squares in the first two regressions be RSS_1 and RSS_2, respectively and that in the case of multiple linear regression be RSS_{12}. Show that $RSS_{12} \leq \min\{RSS_1, RSS_2\}$.

9. Consider two sets of independent observations on (X, Y). Let $\hat{\beta}_1$ be the least-squares estimate of the regression coefficient β computed from the first set and $\hat{\beta}_2$ be that computed from the second data set.

(a) If $\hat{\beta}_1$ and $\hat{\beta}_2$ both are positive, can the estimate of the regression coefficient computed from the pooled data be negative? Justify your answer.

(b) If the means of X observations are equal in these data sets, will you modify your answer?

10. Consider a linear regression model $Y_{n\times 1} = \mathbf{X}_{n\times 2}\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix} + \epsilon_{n\times 1}$ with two non-stochastic covariates and no intercept, where $E(\epsilon) = \mathbf{0}$ and Disp(ϵ)=$\sigma^2\mathbf{I}$. Assume that the values of all three variables X_1, X_2 and Y are standardised (i.e, they have sample mean 0 and sample variance 1). Let r_i be the sample correlation between X_i and Y $(i = 1, 2)$ and r be the sample correlation between X_1 and X_2. If $\widehat{\beta}_1 > \widehat{\beta}_2 > 0$ are the β_1 and β_2, respectively, comment on the validity of the following statements.

(a) The sample correlations r_1 and r_2 are positive.

(b) r_1 is larger than r_2.

(c) The ridge regression estimates of β_1 and β_2 are positive (you can give an algebraic or a geometric argument).

(d) For any $\ell = (\ell_1, \ell_2)$ with $\|\ell\| = 1$, the variance of $\ell_1\widehat{\beta}_1 + \ell_2\widehat{\beta}_2$ cannot be smaller than $\sigma^2/n(1 + |r|)$.

11. (a) Consider a multiple linear regression problem, where the design matrix $\mathbf{X}$ is such that $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$, the identity matrix. Show that the LASSO estimates of the parameters of this multiple linear regression model can be obtained by using a smoothing operator S_λ on the coordinates of the usual least squares estimate, where S_λ is defined as

$$S_\lambda(t) = \begin{cases} t - \frac{\lambda}{2}\,\text{sign}(t) & \text{if } |t| > \frac{\lambda}{2} \\ 0 & \text{otherwise.} \end{cases}$$

(b) Discuss how the coordinate descent method can be used for computing the LASSO estimates of the model parameters when $\mathbf{X}^\top\mathbf{X} \neq \mathbf{I}$.

12. Consider a linear model $Y_i = \beta_0 + \beta_1 x_{1i} + \ldots + \beta_p x_{pi} + \epsilon_i \quad$ for $i = 1, 2, \ldots, n$, where the predictor variables are non-stochatic and ϵ_i's are iid G, for G being a univaraiate distribution mean 0 and variance σ^2.

(a) If G is normal, show that the maximum likelihood estimate of $(\beta_0, \ldots, \beta_p)$ coincides with the least square estimate.

(b) If G is double exponential, show that the maximum likelihood estimate of $(\beta_0, \ldots, \beta_p)$ coincides with the LAD estimate.

(c) If G is normal, show that ridge and LASSO estimates of $(\beta_0, \ldots, \beta_p)$ can be viewed as the MAP estimates (maximum a posteriori estimate used in Bayesian analysis) for suitable choices of the prior distributions.

13. (a) The sub-gradient of a function $f : \mathbb{R} \to \mathbb{R}$ at a point x_0 is defined as the set $\partial f(x)|_{x=x_0} = \{c \in \mathbb{R} : f(x) - f(x_0) \geq c(x - x_0) \text{ for all } x \in \mathbb{R}\}$. Show that 0 belongs to the set if and only if x_0 is a minimiser of f.

(b) Show that if f is a convex function and the derivative $f'(x_0)$ exists at x_0, then $\partial f(x)|_{x=x_0} = \{f'(x_0)\}$.

(c) Consider two functions f_1 and f_2. Define $f = f_1 + f_2$. Show that if $c_1 \in \partial f_1(x)|_{x=x_0}$ and $c_2 \in \partial f_2(x)|_{x=x_0}$, then $c_1 + c_2 \in \partial f(x)|_{x=x_0}$

14. The sub-gradient of a function $f : \mathbb{R}^p \to \mathbb{R}$ at a point $\mathbf{x}_0$ is defined as the set $\partial f(\mathbf{x})|_{\mathbf{x}=\mathbf{x}_0} = \{\mathbf{c} \in \mathbb{R}^p : f(\mathbf{x}) - f(\mathbf{x}_0) \geq \mathbf{c}^\top(\mathbf{x} - \mathbf{x}_0) \text{ for all } \mathbf{x} \in \mathbb{R}^p\}$. Show that $\mathbf{0} = (0, 0, \ldots, 0) \in \mathbb{R}^p$ belongs to the set if and only if $\mathbf{x}_0$ is a minimiser of f.

15. (a) Find the sub-gradient of the function $f(\mathbf{x}) = \|\mathbf{x}\|_1$ at $\mathbf{x}_0$ when (i) $\mathbf{x}_0 = \mathbf{0}$ and (ii) $\mathbf{x}_0 \neq \mathbf{0}$, where $\|\mathbf{x}\|_1$ denotes the ℓ_1-norm of $\mathbf{x}$.

    (b) Find the sub-gradient of the function $f(\mathbf{x}) = \|\mathbf{x}\|_2$ at $\mathbf{x}_0$ when (i) $\mathbf{x}_0 = \mathbf{0}$ and (ii) $\mathbf{x}_0 \neq \mathbf{0}$, where $\|\mathbf{x}\|_2$ denotes the ℓ_2-norm (i.e., the Euclidean norm) of $\mathbf{x}$.

16. (a) Describe the coordinate descent algorithm for elastic net regression.

    (b) Give an example of a regression problem where the LASSO solution is not unique but the elastic net solution is unique.

17. Consider a regression problem with two covariates, where $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$, the identity matrix. Suppose that all variables are centred, and the least squares fit of Y on X_1 and X_2 is given by $Y = 0.9X_1 - 0.6X_2$. Find the minimizer of the following problems.

    (a) Minimize $\sum_{i=1}^n [Y_i - \beta_1 X_{1i} - \beta_2 X_{2i}]^2$ subject to $\beta_1^2 + \beta_2^2 = 1$.

    (b) Minimize $\sum_{i=1}^n [Y_i - \beta_1 X_{1i} - \beta_2 X_{2i}]^2$ subject to $|\beta_1| + |\beta_2| = 1$.

18. (a) Let $t_0 < t_1 < t_2 < \ldots < t_{k+1}$ be $k + 2$ many real numbers. Consider a class of functions which are defined on $[t_0, t_{k+1}]$, continuous on (t_0, t_{k+1}) and linear in $[t_i, t_{i+1}]$ for each $i = 0, 1, 2, \ldots, k$. Show that any function in this class can be uniquely written as a linear combination of the functions 1, x, $(x - t_1)_+$, $(x - t_2)_+$,..., $(x - t_k)_+$.

    (b) Consider a data set $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$. If there are no observations (no x-values) in the interval (t_4, t_6), show that the design matrix for fitting linear splines with truncated power basis functions will not be of full column rank.

19. (a) Show that various statistical models can be obtained from the perceptron model for suitable choices of the network architecture and transformation functions.

    (b) Show that a classifier based on a single-layer perceptron model is always a linear classifier.

    (c) Consider a regression problem with the true regression function $f(x_1, x_2) = x_1 x_2$. Show that this true regression function can be obtained from a neural network (perceptron) model for suitable choices of the network architecture, edge weights and transformation functions.

20. (a) Show that the Nadaraya-Watson estimator of a regression function can be viewed as a local constant estimator.

    (b) Consider a Nadaraya-Watson estimator of a regression function based on a Gaussian kernel. Discuss the behaviour of this estimator for very small and very large values of the bandwidth.

    (c) Discuss how one can construct a local linear estimator of a regression function using a Gaussian kernel. Show that it turns out to be the usual linear regression estimator for large values of the bandwidth.

21. Let $T_0 \supset T_1 \supset \cdots \supset T_k$ be a nested sequence of regression trees. Consider the cost function

$$\mathrm{RSS}_\alpha(T) = \mathrm{RSS}(T) + \alpha|\widetilde{T}|,$$

where $\mathrm{RSS}(T)$ denotes the residual sum of squares and $|\widetilde{T}|$ denotes the cardinality of the set of leaf nodes in the tree T. For $i = 1, 2, \ldots, k$, define $\eta_i = \Big(\mathrm{RSS}(T_i) - \mathrm{RSS}(T_0)\Big)/\Big(|\widetilde{T}_0| - |\widetilde{T}_i|\Big)$. If $\eta_1 > \eta_2$, show that, for all choices of $\alpha \geq \eta_2$, $\mathrm{RSS}_\alpha(T_1)$ exceeds $\mathrm{RSS}_\alpha(T_2)$.

22. Consider a J-class classification problem. If a node t of a classification tree contains a proportion p_j of observations from the jth class, $j = 1, 2, \ldots, J$, its impurity function is defined as

$$i(t) = \psi(p_1, \ldots, p_J) = \sum_{i \neq j} p_i p_j = 1 - \sum_{j=1}^{J} p_j^2.$$

(a) Show that ψ is a concave function of its arguments. Hence show that ψ is maximized when $p_j = \frac{1}{J}$ for $j = 1, 2, \ldots, J$, and it is minimized when $\max_{1 \leq j \leq J} p_j = 1$.

(b) Show that $\psi(p_1, p_2, \ldots, p_J) \geq \frac{1}{2}\left(1 - \max_{1 \leq j \leq J} p_j\right)$. When does equality hold?

(c) Prove or disprove the following statement:

> "Since ψ is concave, for any split of a node t, the reduction in the impurity function, $\Delta i(t)$, is non-negative."

23. In order to construct a classification tree for a two-class classification problem, one needs to find the best split based on each of the measurement variables. Consider a categorical measurement variable C, which has three categories C_1, C_2, and C_3. The numbers of observations in these three categories are given below:

Category	C_1	C_2	C_3	Total
No. of observations from Class 1	150	200	150	500
No. of observations from Class 2	0	350	150	500
Total	150	550	300	1000

If the impurity function $\psi(p_1, p_2)$ is concave and symmetric in its arguments, find the best split based on the measurement variable C.

24. Give an example of a classification problem where the distributions of the two classes are absolutely continuous but the Bayes classifier is not unique.

25. Show that in a two-class classification problem, the Bayes risk can never exceed the smaller of the two prior probabilities. Give an example of a two-class classification problem where the densities of the two competing classes are not identical but the Bayes risk attains this upper bound.

26. Consider a k-class classification problem, where $\pi_1, \pi_2, \ldots, \pi_k$ are the prior probabilities of the k competing classes. Show that the Bayes risk cannot exceed $1 - (\pi_1^2 + \pi_2^2 + \cdots + \pi_k^2)$. Is it possible to attain this upper bound? Justify your answer.

27. Consider a two-class classification problem between two bivariate normal distributions

$$N_2(0, 0, 1, 1, \rho) \quad \text{and} \quad N_2(0, 0, 1, 1, -\rho),$$

where $\rho > 0$. Assuming equal prior probabilities for the two classes, find the Bayes classifier. Compute the Bayes risk and show that it is a decreasing function of ρ.

28. Consider a two-class classification problem in $\mathbb{R}^d$, where one class follows the standard normal distribution and the other one follows a truncated standard normal distribution with support

$$\left\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| < 10\right\}.$$

Assume that the two classes have the same prior probability 1/2. Find the Bayes rule and the corresponding Bayes risk. Show that the Bayes risk shrinks to zero as d increases.

29. Consider a classification problem where the densities of the two competing classes are given by

$$f_1(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \text{ and } f_2(x) = \begin{cases} 1 + \sin(2k\pi x), & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

where k is a positive integer (so that f_2 is a valid density on $[0, 1]$). Assuming that the prior probabilities of the two classes are equal, find the Bayes classifier and check whether the corresponding Bayes risk depends on the constant k.

30. Consider a two-class classification problem where each of the competing classes is an equal mixture of two bivariate normal distributions. While one class is a mixture of

$$N_2\left(1, 1, \frac{1}{4}, \frac{1}{4}, 0\right) \quad \text{and} \quad N_2\left(-1, -1, \frac{1}{4}, \frac{1}{4}, 0\right),$$

the other one is a mixture of

$$N_2\left(-1, 1, \frac{1}{4}, \frac{1}{4}, 0\right) \quad \text{and} \quad N_2\left(1, -1, \frac{1}{4}, \frac{1}{4}, 0\right).$$

If the two classes have the same prior probability, find the Bayes classifier and the corresponding Bayes risk.

31. Consider a two-class classification problem where one class is $N_2(0, 0, 1, 1, 0)$, and the measurement vector in the other class is distributed as

$$(Y_1 = X_1 \operatorname{sign}(X_2), \quad Y_2 = X_2 \operatorname{sign}(X_1)),$$

where X_1 and X_2 are i.i.d. $N(0, 1)$. If the prior probabilities of the two classes are equal, show that the Bayes risk is 1/4.

32. If $X \sim N(\mu_1, \Sigma)$ and $Y \sim N(\mu_2, \Sigma)$,, where X and Y are independent, show that $P\{\alpha(X-Y) > 0\}$ is maximized when $\alpha \propto \Sigma^{-1}(\mu_1 - \mu_2)$..

33. Consider a classification problem between two univariate distributions with probability density functions

$$f_1(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \text{ and } f_2(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Choose the prior probabilities in such a way that the two types of misclassification probabilities for the corresponding Bayes classifier are equal.

34. Consider a two-class classification problem where the densities of the two classes are given by

$$f_1(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\} \quad \text{and} \quad f_2(x) = \frac{1}{2} \exp\{-|x|\}, \text{ respectively.}$$

(a) Find the maximum value of π_1 (the prior probability of the first class) for which the Bayes classifier classifies all observations into the second class.

(b) Can you choose a value of $\pi_1 < 1$ such that the Bayes classifier classifies all observations into the first class? Justify your answer.

35. A student was asked to generate one observation from the standard normal distribution and one observation from the standard double exponential distribution (whose densities are given in the previous question). The student generated the two observations

$$0.8917 \quad \text{and} \quad -0.2643,$$

but forgot which observation was generated from the standard normal distribution. Describe how you would help the student label these observations.

36. Comment on the validity of the statement: "In a binary classification problem with equal prior probabilities for the two classes, if two competing classes differ only in their scales, any linear classifier will have average misclassification probability 0.5".

37. Consider a classification problem between two multivariate normal distributions $N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. If the two classes have equal prior probabilities, show that (i) the Bayes classifier is linear and (ii) the Bayes risk is a decreasing function of $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

38. Consider a classification problem with equal priors, where the two distributions have densities

$$f_1(\mathbf{x}) = C|\boldsymbol{\Sigma}|^{-1/2}\psi\Big((\mathbf{x}-\boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_1)\Big) \text{ and } f_2(\mathbf{x}) = C|\boldsymbol{\Sigma}|^{-1/2}\psi\Big((\mathbf{x}-\boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_2)\Big),$$

respectively. If $\psi : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly decreasing function, show that the Bayes classifier is linear. Will the Bayes classifier remain linear if the priors are not equal? Justify your answer.

39. In a binary classification problem between two d-dimensional normal distributions $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $N(\boldsymbol{\mu}, 4\boldsymbol{\Sigma})$, show that the Bayes risk can be expressed in the form $a_0 + a_1 G_d(b_1) + a_2 G_d(b_2)$, where $G_d(\cdot)$ is the cdf of the chi-square distrbution with d degrees of freedom. Compute a_0, a_1, a_2, b_1 and b_2 in terms of prior probabilities and parameters of the two classes.

40. Show that for a classification problem with equal prior probabilities, the Bayes risk is $1/2$ if and only if the two distributions are identical.

41. Prove or disprove the following statement: "In a classification problem with equal prior probabilities, if the two multivariate distributions are different (even if they do not differ in their locations), there must be a linear classifier which has misclassification probability smaller than 0.5".

42. If $X_1, X_2, \ldots, X_d$ are i.i.d. $N(0,1)$, $S = \sum_{i=1}^d X_i^2$ follows the chi-square distribution with d degrees of freedom. Using this result, find the volume of the d-dimensional unit ball and the surface area of the d-dimensional unit sphere.

43. Let $\hat{f}_h$ be the histogram estimator of a density function based on a common bin-width h. If the underlying density function f has a bounded derivative, and $h = h_n = O(n^{-\delta})$ for some $\delta \in (0,1)$ (so that $h \to 0$ and $nh \to \infty$), show that for any fixed x, $\hat{f}_h(x)$ converges to $f(x)$ in probability as the sample size n grows to infinity.

44. Let $X_1, X_2, \ldots, X_n$ be independent copies of a random variable X, which follows a distribution with density f. If $\hat{f}_h$ is the kernel density estimator of f based on a symmetric kernel with bandwidth h, where f is bounded and has a bounded, continuous second derivative and the kernel is a bounded symmetric density with finite second moment, show that for any given x, $\text{Bias}(\hat{f}_h(x)) = O(h^2)$ and $\text{Var}(\hat{f}_h(x)) = O(\frac{1}{nh})$ as $h \to 0$.

45. Consider a binary classification problem with π and $1-\pi$ as the prior probabilities of the two classes and we have 10 bivariate observations from each of these classes. Let $\delta_h(\cdot)$ be the kernel discriminant analysis rule constructed based on the Gaussian kernel with bandwidth h. Draw a rough sketch of the class boundary when (i) $\pi = 0.4$, $h \downarrow 0$, (ii) $\pi = 0.4$, $h \uparrow \infty$, (iii) $\pi = 0.5$, $h \downarrow 0$, (iv) $\pi = 0.5$, $h \uparrow \infty$.

46. Describe what is the difference between bagged regression/classification and a random forest method. What is an out-of-bag estimate of error.

47. Let $\mathbf{0}_d$ be the origin in the d-dimensional space, and D_n be the distance between the origin and its nearest neighbor in a set of n independent observations generated from the d-variate standard normal distribution. Show that
(i) For any fixed $\epsilon > 0$, $P(D_n > \epsilon) \to 0$ if d is fixed and n tends to infinity.
(ii) For any fixed $M < \infty$, $P(D_n > M) \to 1$ if n is fixed and d tends to infinity.

48. Consider a classification problem between two distributions, one of which in uniform on $\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le a\}$ and the other one is uniform on $\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le b\}$. If $a \neq b$, show that for any fixed prior probabilities of the two classes, the error rate of the Bayes classifier converges to 0 as d grows to infinity.

49. Show that the integral of the k-nearest neighbor density estimate is not finite in one dimension. Describe how one get the nearest neighbor classifier from the idea of nearest neighbor density estimation.

50. Show that the classifier obtained from a single-layer perceptron model is a linear classifier. Show that minimising the softmax loss function for a single-layer perceptron model is equivalent to maximising the likelihood function in logistic regression.

51. Describe the logistic regression method for a classification problem involving more than two classes. What is the GLMNET classifier?

52. Describe how you will use a multi-layer perceptron model for a classification problem involving more than two classes. Will it lead to nonlinear class boundaries?

53. A data set contains the information on the age (X_1), gender (X_2), typing speed (X_3), concentration level (X_4), daily working hours (X_5) and average no of mistakes (Y) on a page for different typists. Use an appropriate statistical model for this data set for predicting the average number of mistakes based on other covariates.

54. Consider a regression problem, where the response variable Y takes only two values, 0 and 1, and they are arranged below according to increasing values of a covariate X_1.

X_1	0.6	1.4	2.2	2.5	2.9	3.2	3.8	4.7	5.3	5.6	6.5	7.0	8.6	9.4	10.1
Y	0	0	0	0	0	1	0	1	0	1	0	1	1	1	1

Find the best split based on X_1 using the Gini index as the impurity function. Does your answer change if the entropy or the misclassification rate is used instead?

55. Let $\widehat{f}_{k,n}$ be the k-th nearest-neighbor estimate of a regression function based on n ($n > 10$) observations with distinct x-values. If $\widehat{f}_{1,n}(x) \leq 1$for all $x \in \mathbb{R}^d$, show that $\widehat{f}_{2,n}(x) \leq 1$for all $x \in \mathbb{R}^d$. Is the converse true? Justify your answer.

56. Consider the classification problem given in Question no 30. Give an ideal perceptron model (with appropriate network architecture and transformation function) for this problem.

57. Consider the same problem (given in Question no. 30). Suppose that we have some observations from each of the two classes. Now, a person assumes the underlying distributions for both classes to be normal and carries out quadratic discriminant analysis. How will the classifier perform on this data set? Justify your answer.

58. Describe how the Nadaraya-Watson estimator of a regression function can be meaningfully used for a binary classification problem. How can it be used for classification problems involving more than two classes?

59. The central region of the d-dimensional standard normal distribution with density ϕ_d is defined as $\mathcal{C}_d = \{\mathbf{x} : \ \phi_d(\mathbf{x})/\phi_d(\mathbf{0}) > 0.01\}$. Show that if $\mathbf{X}_d$ denotes a d-dimensional standard normal variate, $P(\mathbf{X}_d \in \mathcal{C}_d) \to 0$ as d grows to infinity.

60. Give an example of a binary classification problem with equal priors, where the usual nearest neighbor classifier classifies almost all future observations to a single class. Justify your answer.

61. Consider a classification problem involving observations from two 100-dimensional absolutely continuous probability distributions. If there are 40 observations from the first class and 50 observations from the second class, show that with probability one, there will be a linear classifier which will perfectly separate the observations from the two classes.

62. Consider the setup considered in Question no. 61. Show that in this case, the maximum likelihood estimates of the parameters of the logistic regression model does not exist.

63. Consider a classification problem involving J classes. If the observations are d-dimensional, how many parameters are there in the logistic regression model? Suppose that you know how to fit a generalized linear model with logit link function. How will you modify your method to fit a generalized additive model with the logit link?

64. What is probit regression? Describe the iteratively reweighted least squares method for fitting a probit regression model for a binary classification problem.

65. For each of the following models, determine whether it is a Generalized Additive Model (GAM). Justify your answers by identifying an appropriate link function, where applicable.
    1. $E(Y \mid X_1, X_2) = e^{X_1 + 2X_2}$.
    2. $E(Y \mid X_1, X_2) = e^{X_1 X_2}$.
    3. $\log E(Y \mid X_1, X_2) = \sin(X_1) + X_2^2$.
    4. $\log \dfrac{P(Y = 1 \mid X_1, X_2)}{1 - P(Y = 1 \mid X_1, X_2)} = X_1^2 + \log(1 + X_2^2)$.
    5. $P(Y = 1 \mid X_1, X_2) = \dfrac{1}{1 + e^{-(X_1 + X_2^2)}}$.
    6. $\log E(Y \mid X_1, X_2) = X_1^2 + X_2^2 + X_1 X_2$.

Assume that the conditional distribution of Y belongs to an appropriate exponential-family distribution, and that the displayed link functions are valid for the response under consideration.

66. Formulate the optimization problem for constructing the linear SVM classifier when the observations from the two classes are linearly separable. With the help of the Lagrangian, construct the corresponding dual problem. If the solution of the dual problem is known, how will you classify a future observation? How will you modify these steps if the observations from the two classes are not linearly separable?

67. Let $x = (x_1, x_2)^\top$ and $y = (y_1, y_2)^\top$ be arbitrary observations in $\mathbb{R}^2$. For each of the following functions $K_i : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, determine whether K_i is a reproducing kernel of some Hilbert space of functions on $\mathbb{R}^2$. Justify your answers.

    1. $K_1(x, y) = x_1y_1 + x_2y_2$.
    2. $K_2(x, y) = x_1^2y_1^2 + 2x_1x_2y_1y_2 + x_2^2y_2^2$.
    3. $K_3(x, y) = x_1^2y_1^2 + x_2y_2$.
    4. $K_4(x, y) = 1 + x_1y_1 + x_2y_2$.
    5. $K_5(x, y) = x_1y_1 + x_2y_2 + x_1x_2y_1y_2$.
    6. $K_6(x, y) = x_1^2y_1^2 - x_2^2y_2^2$.
    7. $K_7(x, y) = x_1y_1 + x_2y_2 + x_1y_2$.
    8. $K_8(x, y) = \sin x_1 \sin y_1 + \cos x_1 \cos y_1 + \sin x_2 \sin y_2 + \cos x_2 \cos y_2$.
    9. $K_9(x, y) = e^{x_1+y_1} + e^{x_2+y_2}$.
    10. $K_{10}(x, y) = x_1^2y_1^2 + x_2^2y_2^2 + 1$.

68. Suppose we are given a binary classification problem with training observations (x_i, y_i), $i = 1, \ldots, n$, where $x_i \in \mathbb{R}^2$ and $y_i \in \{-1, +1\}$. The two classes are not linearly separable in the original input space.

    1. Explain how a nonlinear SVM can be constructed using a feature map $\phi : \mathbb{R}^2 \to \mathcal{H}$, where $\mathcal{H}$ is a higher-dimensional feature space.

    2. Derive the dual optimization problem for a soft-margin SVM using a kernel function $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle_{\mathcal{H}}$.

    3. Consider the Gaussian radial basis function (RBF) kernel

    $$K(x, z) = \exp\left(-\gamma \|x - z\|^2\right), \qquad \gamma > 0.$$

    Explain why this kernel allows nonlinear decision boundaries.

    4. Given the optimal dual coefficients $\alpha_1^*, \ldots, \alpha_n^*$ and intercept b^*, write down the decision function for classifying a new observation x.

    5. Explain how the support vectors are identified from the optimal dual coefficients. Distinguish between support vectors lying on the margin and those lying inside the margin or being misclassified.

69. For the classification problems in Question no. 30 and Question no. 31, what type of kernel will you recommend for classification using nonlinear SVM? Justify your answer.

70. What is the main idea of boosting? What are its advantages and disadvantages over bagging? Describe the AdaBoost and Gradient Boosting algorithms for regression and classification. What are the main differences between Gradient Boosting and XGBoost?